Coding & Solution

Date	15 March 2026
Team ID	SWTID-2026-6912
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented FitBuddy solution. The application uses FastAPI with Google Gemini models to generate personalized 7-day workout plans, nutrition/recovery tips, and feedback-based plan updates, with SQLite/SQLAlchemy persistence and a Jinja2 web interface.

Instructions:

1. Ensure the code is modular, readable, and follows good development practices.

2. The solution should address the core fitness-planning problem and implement the key generation, feedback, storage, and admin-view features.

3. Include source files, configuration files, dependencies, and setup instructions needed to run the application.

4. Attach or provide the code repository link when available.

Solution Summary

Field	Details
Repository Link / URL	To be updated / provided by the team
Programming Language(s)	Python, HTML, CSS, Jinja2
Framework(s) Used	FastAPI, SQLAlchemy; Google Gemini API integration
Key Features Implemented	Personalized 7-day workout generation; Gemini Pro workout generation and feedback updates; Gemini Flash nutrition/recovery tips; SQLite user/plan storage; admin view; Jinja2 templates; FastAPI API endpoints.
Pending / Incomplete Features	Cloud deployment is not covered in the current project documentation; repository URL is not provided.
Setup / Run Instructions	Create/activate a virtual environment; install dependencies with **pip install -r requirements.txt** (or the listed FastAPI/Uvicorn/Jinja2/SQLAlchemy/Google Generative AI packages); configure **GOOGLE_API_KEY**; run with Uvicorn; open **http://127.0.0.1:8000** and **/docs**.

Team Work Allocation

S.No	Team Member	Assigned Work
1	Pavithra M	**Milestone 1 – Model Selection & Architecture:** research/select Gemini models, define application architecture, and set up the development environment.
2	Manisha S	**Milestone 2 – Core Functionalities:** develop workout generation, nutrition/recovery tips, feedback-based plan updating, database storage, and FastAPI input processing.
3	Monica G	**Milestone 3 – routes.py Development:** implement the main application routes, connect AI functions, process requests, store plans, and support the admin view.
4	Monisha Anandan	**Milestone 4 – Frontend Development:** design responsive HTML/CSS pages, Jinja2 templates, forms, result display, feedback interface, and admin page.
5	Mridula M	**Milestone 5 – Deployment & Testing:** prepare local deployment, configure environment variables/dependencies, run Uvicorn, test pages and API documentation.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	To be updated

Additional Notes / Comments:

• FitBuddy generates personalized 7-day workout plans from user details such as age, weight, fitness goal, and workout intensity.
• Gemini 1.5 Pro is documented for structured workout generation and feedback-based plan updates; Gemini Flash is used for quick nutrition/recovery tips.
• SQLite with SQLAlchemy stores user information and both original and updated plans.
• The current documentation describes local deployment; cloud deployment and the final repository URL can be added later.

Project Technology Snapshot

Layer	Technology / Component	Purpose
Frontend	HTML, CSS, Jinja2	User input, plan display, feedback form, admin interface
Backend	FastAPI, Python	Routing, request handling, API endpoints
AI	Google Gemini Pro & Flash	Workout generation, plan updates, nutrition/recovery tips
Database	SQLite + SQLAlchemy	Persistent storage of users and workout plans
Server	Uvicorn	Local ASGI application server